

File Descriptors

Program 1

```
himasree@LAPTOP-41COT5AA:~$ cd ~/OSLab
himasree@LAPTOP-41COT5AA:~/OSLab$ nano std.c
himasree@LAPTOP-41COT5AA:~/OSLab$ mkdir file_D
himasree@LAPTOP-41COT5AA:~/OSLab$ cd file_D
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ program1.c
program1.c: command not found
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ nano program1.c
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc program1.c -o program1
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./program1
Hello World
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ |
```

```
#include <unistd.h>
int main()
{
    write(STDOUT_FILENO, "Hello World\n", 12);
    return 0;
}
```

Program 2

```
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ nano program2.c
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc program2.c -o program2
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./program2
1
1
```

```
#include <unistd.h>
int main()
{
    char buffer[20];
    read(STDIN_FILENO, buffer, sizeof(buffer));
    write(STDOUT_FILENO, buffer, sizeof(buffer));
    return 0;
}
```

Program 3

```
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ nano program3.c
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc program3.c -o program3
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./program3
hello
hello
Enter something: himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./program3
```

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    char buffer[100];
    int n;
    printf("Enter something: ");
    n = read(STDIN_FILENO, buffer, sizeof(buffer) - 1);
    if (n > 0)
    {
        buffer[n] = '\0';
        write(STDOUT_FILENO, buffer, n);
    }
    return 0;
}
```

Program 4

```
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ nano program4.c
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc program.4 -o program4
/usr/bin/ld: cannot find program.4: No such file or directory
collect2: error: ld returned 1 exit status
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc program4.c -o program4
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./program4
This is an error
```

```
#include <unistd.h>
int main()
{
    write(STDERR_FILENO, "This is an error\n", 17);
    return 0;
}
```

Program 5

```
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ nano program5.c
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc program5.c -o program5
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./program5
open: No such file or directory
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int fd;
    fd = open("doesnotexist.txt", O_RDONLY);
    if (fd == -1)
    {
        perror("open");
        return 1;
    }
    close(fd);
    return 0;
}
```

Program 6

```
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ nano program6.c
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc program6.c -o program6
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./program6
File created successfully!
File descriptor = 3
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int fd;
    fd = open("student.txt", O_CREAT | O_WRONLY, 0644);
    if (fd == -1)
    {
        perror("open");
        return 1;
    }
    printf("File created successfully!\n");
    printf("File descriptor = %d\n", fd);
    close(fd);
    return 0;
}
```

Program 7

```
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ nano program7.c
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc program7.c -o program7
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./program7
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ |
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int fd;
    fd = open("student.txt", O_WRONLY | O_TRUNC);
    if (fd == -1)
    {
        perror("open");
        return 1;
    }
    write(fd, "New Data\n", 9);
    close(fd);
    return 0;
}
```

Dup():

```
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ nano dup.c
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ gcc dup.c -o dup
himasree@LAPTOP-41COT5AA:~/OSLab/file_D$ ./dup
fd1 = 3
fd2 = 4
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int fd1, fd2;
    fd1 = open("student.txt", O_CREAT | O_WRONLY, 0644);
    if (fd1 == -1)
    {
        perror("open");
        return 1;
    }
    printf("fd1 = %d\n", fd1);
    fd2 = dup(fd1);
    printf("fd2 = %d\n", fd2);
    close(fd1);
    close(fd2);
    return 0;
}
```